We are hurting ourselves

Ray Reyes
Kiet Tran
Thanh Nguyen

Thesis & Data

Thesis : Humans Are Bad For The Air Quality

Our thesis explores the idea that human activity harms the environment, especially the air quality. We support this by analyzing data that demonstrates a correlation between population growth, energy consumption and PM2.5 levels.

Density (Simple Maps): https://simplemaps.com/data/us-cities

Air Quality (Environmental Protection Agency): https://www.epa.gov/air-trends/air-quality-cities-and-counties

Pollution emissions (Department of Energy) : https://www.epa.gov/air-trends/air-quality-cities-and-counties

Figure 1

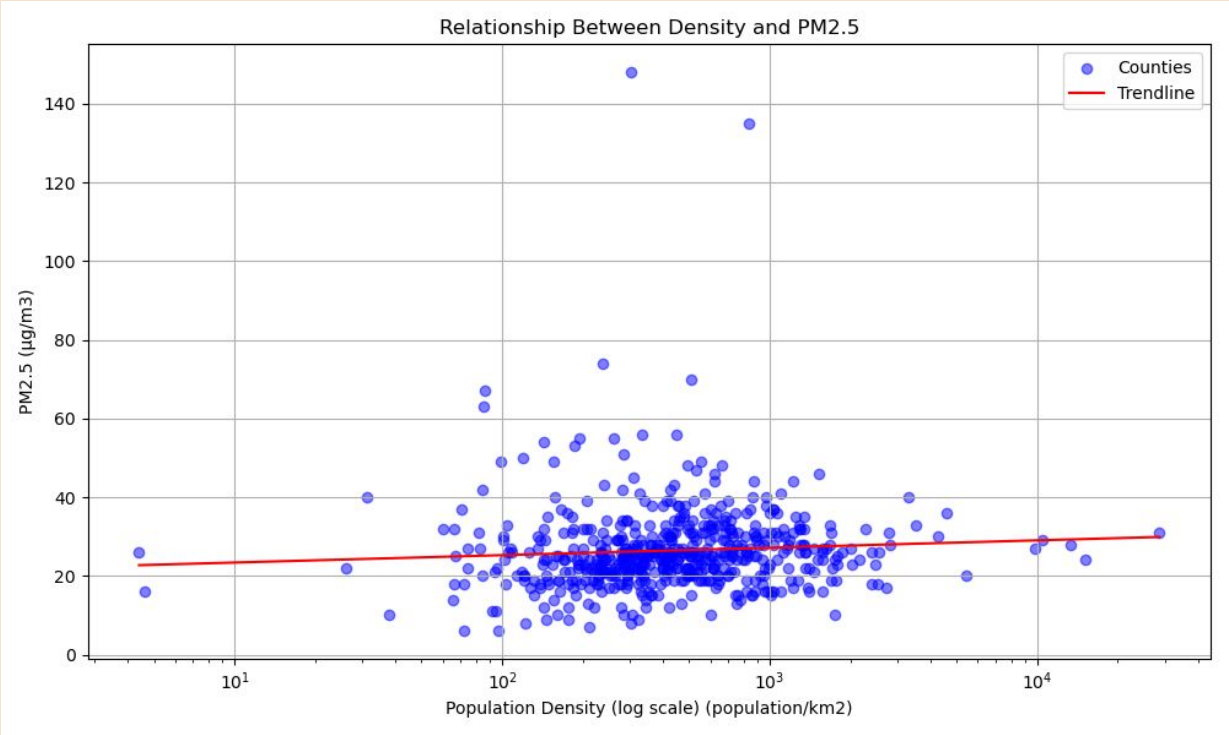

Relationship Between Density and PM2.5
Counties
Trendline
PM2.5 (µg/m3)
0
20
40
60
80
100
120
140
10^1
10^2
10^3
10^4
Population Density (log scale) (population/km2)

Figure 2

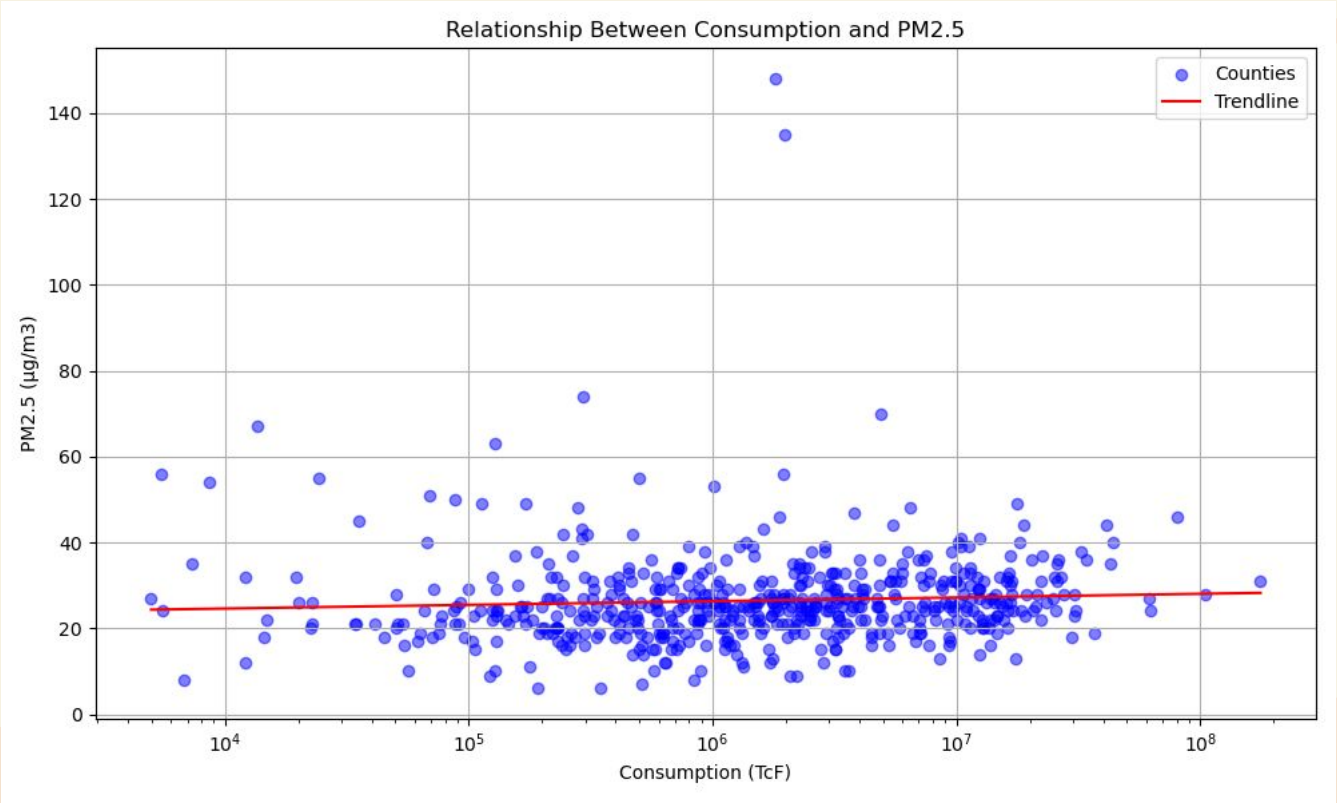
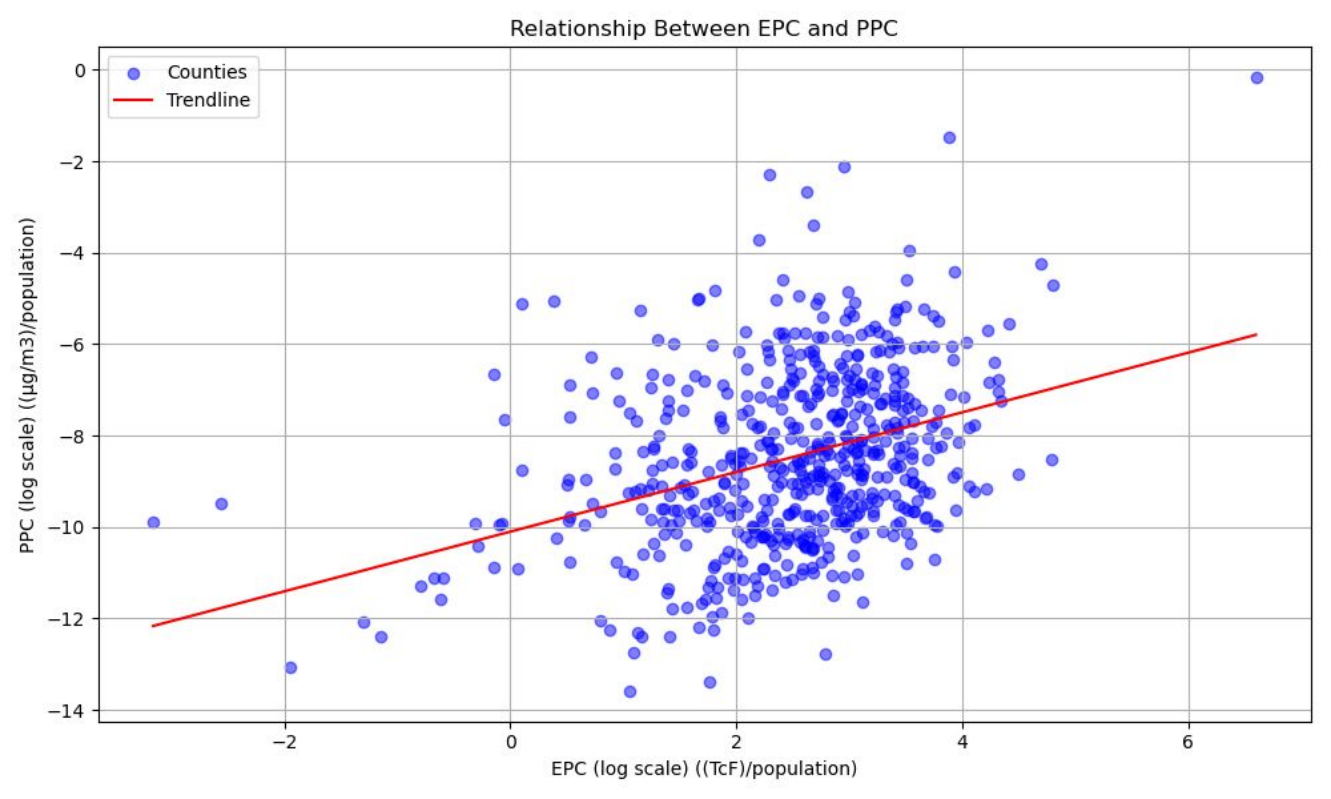

Relationship Between Consumption and PM2.5
Counties
Trendline
PM2.5 (µg/m3)
0
20
40
60
80
100
120
140
10^4
10^5
10^6
10^7
10^8
Consumption (TcF)

Figure 3


Relationship Between EPC and PPC
Counties
Trendline
PPC (log scale) ((µg/m3)/population)
EPC (log scale) ((TcF)/population)
0
−2
−4
−6
−8
−10
−12
−14
−2
0
2
4
6

Conclusion & Implications

Policy:

- **Promote renewable energy and reduce reliance on fossil fuels.**
- **Invest in public transportation and energy-efficient technologies.**
- **Implement stricter air quality standards and monitoring.**

Future direction:

- **Examine industrial and transportation emissions.**

Call to Action:

- **Adopt energy-saving habits to reduce environmental impact.**

OUR ROADMAP

Thesis: Cities with larger populations tend to have poorer air quality, as reflected in the number of unhealthy air quality days.

Reason 1: Show the relationship between population density and PM2.5

Support: Figure 1 shows the higher the population, the higher the PM2.5 (the worse the air quality is)

Reason 2: Show the relationship between energy consumption and PM2.5

Support: Figure 2 shows the higher the energy consumption, the higher the PM2.5 (the worse the air quality is)

Reason 3: Show the relationship between EPC (per capita energy usage) and PPC (per capita pollution)

Support: Figure 3 shows the positive relationship between EPC and PPC